

CLASS -10 (2025-26)

Operators in Java

CHAPTER 4

Assignments:-

A. Tick (✓) the correct answer.

1. The = operator is an example of which operator?

- a. Relational
- b. Logical
- c. Arithmetic
- d. Assignment

Answer:- d. Assignment

2. Which of the following is an example of a postfix increment operator?

- a. a+
- b. a++
- c. ++a
- d. +++a

Answer:- b. a++

3. Which of the following is an example of a prefix decrement operator?

- a. ++
- b. a-
- c. -a
- d. ++a

Answer:- c. -a

4. Which of the following is the correct use of a ternary operator?

- a. (a > b) ? "a is greater than b" : "b is greater than a";
- b. (a > b) "a is greater than b"? "b is greater than a";
- c. Both a & b
- d. (a > b) ? "a is greater than b" : "b is greater than a";

Answer:- d. (a > b) ? "a is greater than b" : "b is greater than a";

5. Which of the following operators can we use to initialize all non-primitive data types?

- a. (dot) operator
- b. Ternary operator
- c. new operator
- d. relational operator

Answer:- c. new operator

B. Fill in the blanks.

1. Arrange these operators (<, ++, +, %) in order of higher precedence to lower precedence

Answer:- ++, %, +, <

2. The equivalent Java expression for $a^2 + b^2 + 2ab$ is

Answer:- a*a + b*b + 2*a*b

3. The output of the expression `a += a++ * ++a % a` is if a is 5.

Answer:- 30

4. The expression `(true) || (true)` will return.....

Answer:- true

5. The expression `10 % 3` will return.....

Answer:- 1

C. Short Answer type question

1. What is the output of the following expressions if executed sequentially? Where $a = 10$, $b = 2$ and $c = 8$.

- `c += a / b * c`

- $a = c + b++ * (a + b + c)$

Answer:-

Step-by-step:

- Initially: $a = 10, b = 2, c = 8$

First expression:

$c += a / b * c$

$\Rightarrow c = c + (a / b * c)$

$\Rightarrow c = 8 + (10 / 2 * 8)$

$\Rightarrow c = 8 + (5 * 8) = 8 + 40 = 48$

Now: $c = 48, b = 2, a = 10$

Second expression:

$a = c + b++ * (a + b + c)$

$\Rightarrow$ Use $b = 2$, but $b++$ means use 2 then increment to 3

$\Rightarrow a = 48 + 2 * (10 + 3 + 48)$

$\Rightarrow a = 48 + 2 * 61 = 48 + 122 = 170$

Final values: $a = 170, b = 3, c = 48$

2. What will be the output of the following ternary expression if $\text{int } a = 5, b = 3$ and double d ?

$d = (a \geq b) ? (a + b) : (a * b);$

Answer:-

$a = 5, b = 3$

$a \geq b$ is true

So, $d = a + b = 5 + 3 = 8$

Answer:- 8.0

3. What is the difference between the ternary operator and the unary operator? Give one example of each.

Answer:-

- **Ternary operator** evaluates a condition and returns one of two values:

Syntax: $(\text{condition}) ? \text{value_if_true} : \text{value_if_false}$

Example: $\text{int max} = (a > b) ? a : b;$

- **Unary operator** operates on a single operand to change its value.

Example: $++a;$ (increments a by 1)

4. Write the Java expressions of the following mathematical expressions:

a. $A = (B + C) / 2 \times h$

Answer:- $A = (B + C) / 2 * h;$

b. $V = \pi r^2 \times h$

Answer:- $V = \text{Math.PI} * r * r * h;$

5. Name the operators listed below:

a. $\geq$

Answer:- Relational operator

b. $!=$

Answer:- Relational operator

c. $--$

Answer:- Unary decrement operator

d. $+=$

Answer:- Compound assignment operator

e. $\%$

Answer:- Arithmetic (modulus) operator

f. $\parallel$

Answer:- Logical OR operator

6. Give the output of the following expressions:

a. $a = ++a + a-- / ++a + -a;$ when $a = 2$

Answer:-

Step-by-step:



$a = ++a + a-- / ++a + -a$

Initial $a = 2$

- $++a \rightarrow 3$
- $a-- \rightarrow$ use 3, then becomes 2
- $++a \rightarrow 3$
- $-a \rightarrow -3$

Now expression becomes:

$a = 3 + 3 / 3 + (-3)$

$\Rightarrow a = 3 + 1 + (-3) = 1$

Answer:- 1

b. $i *= j++ \% j-- / k * 10$; when $i = 2, j = 4, k = 3$

Step-by-step:

- $j++ = 4$ (j becomes 5)
- $j-- = 5$ (used, then becomes 4)
- So expression:
 $i *= 4 \% 5 / 3 * 10 = 4 / 3 * 10 = 1 * 10 = 10$
- $i = 2 * 10 = 20$

Answer:- 20

Assertion and Reason type questions

1.

Assertion (A): The expression $a = ++a + a-- / ++a + -a$ gives the output 1 when $a = 2$.

Reason (R): Pre-increment and post-decrement operators affect the variable differently during evaluation.

Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer:- a

2.

Assertion (A): The ternary operator returns one of two values depending on the condition.

Reason (R): The syntax of ternary operator is $(condition) ? value_if_false : value_if_true$.

Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer:- c

3.

Assertion (A): The output of $d = (a >= b) ? (a + b) : (a * b)$ is 8.0 when $a = 5$ and $b = 3$.

Reason (R): Since a is greater than b , the ternary operator evaluates the false branch.

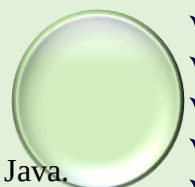
Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer:- c

4.

Assertion (A): The `new` operator is used to allocate memory to non-primitive data types in Java.



Reason (R): The `new` operator is used to create instances of classes and arrays.

Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer:- a

5.

Assertion (A): The expression `10 % 3` evaluates to 1 in Java.

Reason (R): The modulus operator returns the quotient of the division.

Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer:- c

6.

Assertion (A): The expression `(true) || (true)` returns `true`.

Reason (R): The `||` operator returns false only if both conditions are true.

Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer:- c

7.

Assertion (A): The operator `>=` is used to compare two values in Java.

Reason (R): The `>=` operator is an arithmetic operator.

Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer:- c

8.

Assertion (A): The expression `c += a / b * c` updates the value of `c` to 48 when `a = 10`, `b = 2`, `c = 8`.

Reason (R): Operator precedence ensures division and multiplication are performed before addition.

Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

Answer:- a

9.

Assertion (A): `++`, `--`, and `!` are examples of unary operators in Java.

Reason (R): Unary operators operate on a single operand.

Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A



c) A is true but R is false

d) A is false but R is true

Answer:- a

10.

Assertion (A): The Java expression for $V = \pi r^2 \times h$ is `V = Math.PI * r * r * h;`

Reason (R): In Java, π is written as `Math.PI` and exponentiation is performed using `r * r`.

Options:

a) Both A and R are true, and R is the correct explanation of A

b) Both A and R are true, but R is not the correct explanation of A

c) A is true but R is false

d) A is false but R is true

Answer:- a

"What is the output..."

1. What is the output of the following expression if a = 10, b = 2, c = 5?

```
c += a / ++b * c;
```

Answer:-

`b = 3`, so `c = c + (10 / 3 * 5) = 5 + (3 * 5) = 5 + 15 = 20`

2. What is the output of the following if a = 5, b = 2?

```
int x = a++ + ++b;
```

Answer:-

`x = 5 + 3 = 8`, a becomes 6, b becomes 3

3. What is the output of the expression if x = 4, y = 2?

```
int z = x++ * --y + x;
```

Answer:-

`z = 4 * 1 + 5 = 4 + 5 = 9`

4. What is the output if a = 3 and b = 6?

```
int result = a++ + b-- + ++a + --b;
```

Answer:-

`result = 3 + 6 + 5 + 4 = 18`

5. What will be the result if x = 2, y = 3, z = 4?

```
int res = x + y * z % x;
```

Answer:-

`res = 2 + (3 * 4) % 2 = 2 + 12 % 2 = 2 + 0 = 2`

6. What is the output if a = 2, b = 3?

```
int c = ++a * b-- + --b;
```

Answer:-

`c = 3 * 3 + 1 = 9 + 1 = 10`

7. What will be the output if a = 6, b = 4, c = 2?

```
int x = a / b + c * b;
```

Answer:-

`x = 6 / 4 + 2 * 4 = 1 + 8 = 9`

8. What is the output of the following if x = 10, y = 5?

```
int z = x % y + x / y;
```

Answer:-

`z = 10 % 5 + 10 / 5 = 0 + 2 = 2`



9. What is the result if $m = 8$, $n = 3$?

```
int res = m-- - ++n + m;
```

Answer:-

$res = 8 - 4 + 7 = 11$

10. What will be the value of `res` if $a = 2$, $b = 4$, $c = 3$?

```
int res = a * b + c++ / --a;
```

Answer:-

$res = 2 * 4 + 3 / 1 = 8 + 3 = 11$

11. What is the output if $a = 10$, $b = 5$?

```
int result = (a % 2 == 0) ? a / b : b / a;
```

Answer:-

$a \% 2 == 0 \rightarrow \text{true} \rightarrow 10 / 5 = 2$

Result = 2

12. What is the output if $x = 7$, $y = 3$?

```
int res = (x % y > 1) ? x * y : x + y;
```

Answer:-

$7 \% 3 = 1 \rightarrow \text{not greater than 1} \rightarrow \text{false} \rightarrow 7 + 3 = 10$

Result = 10

13. What is the output if $a = 6$, $b = 2$?

```
int result = (a / b == 3) ? a + b : a - b;
```

Answer:-

$6 / 2 = 3 \rightarrow \text{true} \rightarrow 6 + 2 = 8$

Result = 8

14. What is the output if $a = 9$, $b = 4$?

```
int res = a % b + b % a;
```

Answer:-

$9 \% 4 = 1, 4 \% 9 = 4 \rightarrow 1 + 4 = 5$

Result = 5

15. What is the output if $x = 12$, $y = 5$?

```
int result = (x % y == 2) ? x - y : x + y;
```

Answer:-

$12 \% 5 = 2 \rightarrow \text{true} \rightarrow 12 - 5 = 7$

Result = 7

16. What is the output if $a = 15$, $b = 3$?

```
int result = a % b * (a / b);
```

Answer:-

$15 \% 3 = 0, 15 / 3 = 5 \rightarrow 0 * 5 = 0$

Result = 0

17. What is the output if $m = 10$, $n = 4$?

```
int result = (m / n > 2) ? m % n : n % m;
```

Answer:-

$10 / 4 = 2 \rightarrow \text{not greater} \rightarrow \text{false} \rightarrow 4 \% 10 = 4$

Result = 4

18. What is the output if $a = 8$, $b = 2$, $c = 3$?

```
int result = (a % b == 0) ? (a / c) : (a * c);
```

Answer:-



$8 \% 2 = 0 \rightarrow \text{true} \rightarrow 8 / 3 = 2$ (integer division)

Result = 2

19. What is the output if $x = 5$, $y = 2$?

`int res = (x % y == 1) ? x * y : x + y;`

Answer:-

$5 \% 2 = 1 \rightarrow \text{true} \rightarrow 5 * 2 = 10$

Result = 10

20. What is the output if $a = 20$, $b = 7$?

`int result = (a / b % 2 == 0) ? a + b : a - b;`

Answer:-

$20 / 7 = 2, 2 \% 2 = 0 \rightarrow \text{true} \rightarrow 20 + 7 = 27$

Result = 27

21. What is the output if $a = 5$, $b = 2$?

`int result = (++a % b == 0) ? a / b : a * b;`

Answer:-

$++a = 6, 6 \% 2 = 0 \rightarrow \text{true} \rightarrow 6 / 2 = 3$

Result = 3

22. What is the output if $x = 9$, $y = 3$?

`int res = (--x / y > 2) ? x % y : y % x;`

Answer:-

$--x = 8, 8 / 3 = 2 \rightarrow \text{not greater} \rightarrow \text{false} \rightarrow 3 \% 8 = 3$

Result = 3

23. What is the output if $a = 7$, $b = 3$?

`int result = (a++ % --b == 1) ? a + b : a - b;`

Answer:-

$--b = 2, a++ = 7$ (then $a=8$), $7 \% 2 = 1 \rightarrow \text{true} \rightarrow 8 + 2 = 10$

Result = 10

24. What is the output if $m = 6$, $n = 2$?

`int res = (++m / ++n == 3) ? m % n : m + n;`

Answer:-

$++m = 7, ++n = 3, 7 / 3 = 2 \rightarrow \text{false} \rightarrow 7 + 3 = 10$

Result = 10

25. What is the output if $x = 10$, $y = 5$?

`int result = (--x % y == 0) ? x++ / y : x-- * y;`

Answer:-

$--x = 9, 9 \% 5 = 4 \rightarrow \text{false} \rightarrow x-- = 9, x \text{ becomes } 8, 9 * 5 = 45$

Result = 45

26. What is the output if $a = 4$, $b = 2$?

`int res = (++a % b == 1) ? a-- + b : --a * b;`

Answer:-

$++a = 5, 5 \% 2 = 1 \rightarrow \text{true} \rightarrow a-- + b = 5 + 2 = 7$, then $a = 4$

Result = 7

27. What is the output if $a = 3$, $b = 2$?

`int result = (a++ % ++b == 1) ? ++a : --b;`

Answer:-

$a++ = 3$ ($a=4$), $++b = 3, 3 \% 3 = 0 \rightarrow \text{false} \rightarrow --b = 2$

Result = 2



28. What is the output if x = 8, y = 3?

```
int res = (++x / --y < 3) ? x++ : y--;
```

Answer:-

$++x = 9, --y = 2, 9 / 2 = 4 \rightarrow \text{false} \rightarrow y-- = 2, \text{ then } y = 1$

Result = 2

29. What is the output if a = 10, b = 4?

```
int result = (--a / b > 2) ? a-- % b : ++b;
```

Answer:-

$--a = 9, 9 / 4 = 2 \rightarrow \text{not greater} \rightarrow \text{false} \rightarrow ++b = 5$

Result = 5

30. What is the output if m = 6, n = 3?

```
int res = (++m % --n == 1) ? m++ / n : m-- * n;
```

Answer:-

$++m = 7, --n = 2, 7 \% 2 = 1 \rightarrow \text{true} \rightarrow m++ = 7 / 2 = 3, \text{ then } m = 8$

Result = 3

One-word question-answer pairs based on the chapter "Operators in Java":

1. Question: Operator used to assign a value?

Answer: Assignment

2. Question: Operator that checks equality?

Answer: ==

3. Question: Increment operator?

Answer: ++

4. Question: Logical AND operator?

Answer: &&

5. Question: Ternary operator symbol?

Answer: ?:

6. Question: Operator for remainder?

Answer: %

7. Question: Operator used to create objects?

Answer: new

8. Question: Unary operator for negation?

Answer: -

9. Question: Relational operator for not equal?

Answer: !=

10. Question: Logical OR operator?

Answer: ||

Short answer questions

1. Question: What is an operator in Java?

Answer: An operator is a symbol that performs an operation on variables and values.

2. Question: What does the ++ operator do?

Answer: It increments the value of a variable by 1.



3. Question: What is the difference between `==` and `=`?

Answer: `==` is a comparison operator, while `=` is an assignment operator.

4. Question: What type of operator is `?:` in Java?

Answer: It is a ternary operator.

5. Question: Which operator is used to find the remainder?

Answer: The modulus operator `%`.

6. Question: What is the use of the `new` operator in Java?

Answer: It is used to create objects of a class.

7. Question: Name any two relational operators in Java.

Answer: `>`, `<`, `>=`, `<=`, `==`, `!=`

8. Question: What will be the output of `10 / 3` in Java?

Answer: 3 (Integer division)

9. Question: What does the `--` operator do?

Answer: It decrements the value of a variable by 1.

10. Question: Can logical operators be used with non-boolean values in Java?

Answer: No, logical operators like `&&`, `||`, and `!` are used only with boolean values.

11. Question: What type of operator is `+=` in Java?

Answer: It is a compound assignment operator.

12. Question: What is the result of the expression `true || false`?

Answer: `true`

13. Question: What is the precedence of the `/` operator compared to `+`?

Answer: `/` has higher precedence than `+`.



14. Question: Which operator is used to compare two values for inequality?

Answer: !=

15. Question: How many operands does a unary operator work with?

Answer: One operand.

